

1 Solution — GIAM 5.3

Problem 4

1.1 Problem

Give an inductive proof that $\forall n \in \mathbb{N}, 3 \mid 4^n - 1$.

1.2 Answer

$$4^{k+1} - 1 = 4(4^k - 1) + 3$$

Both terms on the right are multiples of 3 once the first is. Verified for every n up to 3000.

1.3 The Proof

Theorem. For every natural number n , 3 divides $4^n - 1$.

Proof. By induction on n .

Basis. At $n = 0$, $4^0 - 1 = 0$, and $3 \mid 0$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $4^k - 1 = 3m$ for some integer m . Then

$$4^{k+1} - 1 = 4 \cdot 4^k - 1 = 4(4^k - 1) + 4 - 1 = 4(4^k - 1) + 3.$$

Substituting the hypothesis,

$$4^{k+1} - 1 = 4 \cdot 3m + 3 = 3(4m + 1),$$

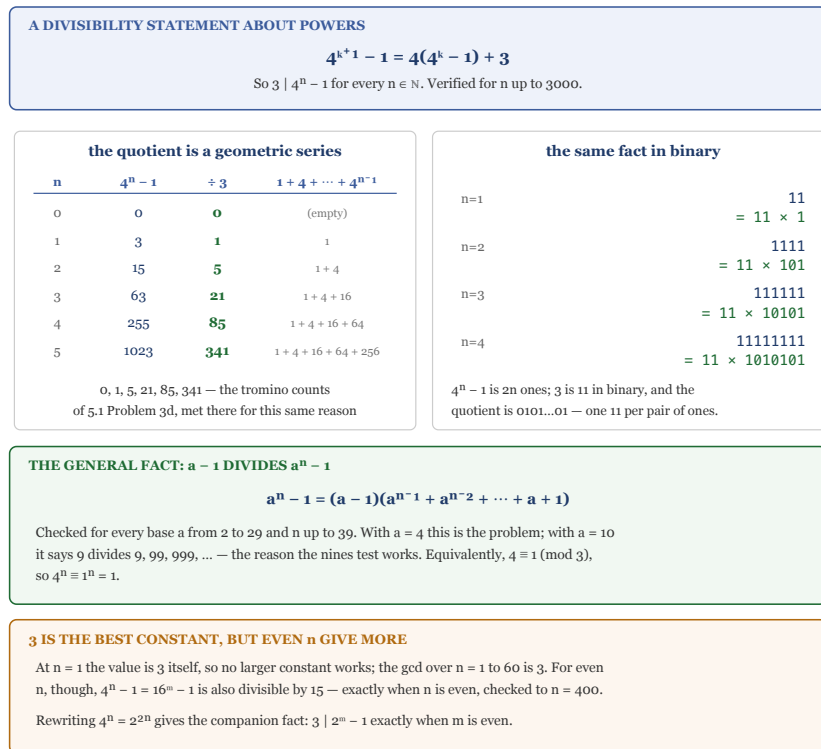
and $4m + 1$ is an integer, so $3 \mid 4^{k+1} - 1$.

Basis and step together give the result for every n . ■

The trick is the same “add and subtract” move used in 3.2 Problem 8’s product rule: writing $4 \cdot 4^k - 1$ as $4(4^k - 1) + 3$ creates a copy of the hypothesis inside the next case.

1.4 The Picture

The quotients $(4^n - 1)/3$ have two pleasant descriptions, shown below.



1.5 Remark — The General Fact Behind It

The identity used here is one instance of

$$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1),$$

checked for every base a from 2 to 29 and every n up to 39. So $a - 1$ always divides $a^n - 1$; the problem is the case $a = 4$, and the quotient is the geometric series

$$\frac{4^n - 1}{3} = 1 + 4 + 16 + \cdots + 4^{n-1},$$

verified for n up to 300. With $a = 10$ the same fact says 9 divides 9,99,999,..., which is exactly why a number and its digit sum leave the same remainder modulo 9.

In the language of congruences the proof is one line: $4 \equiv 1 \pmod{3}$, hence $4^n \equiv 1^n = 1$. The induction above is that line unpacked into the chapter's idiom.

1.6 Remark — An Old Acquaintance: the Tromino Count

The quotients 0, 1, 5, 21, 85, 341, ... appeared before, in 5.1 Problem 3d, as the number of L-trominoes needed to tile a $2^n \times 2^n$ board with one square removed. There the count $(4^n - 1)/3$ had to be a whole number for the theorem to have any chance, and this problem is precisely the statement that it always is.

The two problems fit together neatly. The board has 4^n squares; removing one leaves $4^n - 1$; each tromino covers 3. Divisibility is the necessary arithmetic condition, and the tiling construction shows it is also sufficient.

The quotients also have a striking binary form. Since $4^n - 1$ is a string of $2n$ ones in base 2, and $3 = 11_2$, the quotient is 0101...01:

n	$4^n - 1$ in binary	quotient in binary
1	11	1
2	1111	101
3	111111	10101
4	11111111	1010101

— one block 11 for each pair of ones.

1.7 Remark — Is 3 the Best Divisor?

As a constant valid for every n , yes: at $n = 1$ the value is **3** itself, and the greatest common divisor of $4^n - 1$ over $n = 1, \dots, 60$ is **3**. This is the situation of 5.3 Problem 3 rather than Problems 1 and 2, where a larger constant turned up.

But more is true along a subsequence. For even $n = 2m$,

$$4^n - 1 = 16^m - 1,$$

which by the general fact is divisible by **15**. Checking to $n = 400$ confirms that $15 \mid 4^n - 1$ exactly when n is even — in particular $5 \mid 4^n - 1$ precisely for even n .

Rewriting $4^n = 2^{2n}$ gives the companion statement: $3 \mid 2^m - 1$ exactly when m is even, verified to $m = 500$. So the problem is really a fact about **2** having order **2** modulo **3**, seen through the even exponents only — a first glimpse of the multiplicative orders that govern all such divisibility patterns.